Decentralized Free-Support Wasserstein Barycenter

Xinbao Qiao

Introduction

Wasserstein barycenters average probability measures while preserving the geometry of mass. In decentralized settings, each node sees only a local measure and communicates with neighbors.
Existing decentralized WB methods usually fix a common support grid, reducing the task to consensus over weights (Dvurechensky et al., NeurIPS 2018; Kroshnin et al., ICML 2019). Free support removes that common coordinate system: atom locations move, atom identities become ambiguous, and the optimization is nonconvex.

Theory

Theorem: exact aggregation descends

In a fully connected network,

$$F(\mathbf{B}^{(r+1)}) \le F(\mathbf{B}^{(r)}) - \eta(2-\eta)\Delta^{(r)}.$$

Hence

$$\min_{r<R} \Delta^{(r)} \le \frac{F(\mathbf{B}^{(0)}) - F_m^{\mathrm{inf}}}{R\eta(2-\eta)}.$$

The interpolation step is a relaxed MM update with an $\mathcal{O}(1/R)$ stationarity rate.

Corollary: Sinkhorn variant descends

The entropy-regularized objective obeys the same descent form,

$$F_\tau(\mathbf{B}^{(r+1)}) \le F_\tau(\mathbf{B}^{(r)}) - \eta(2-\eta)\Delta_\tau^{(r)}.$$

Regularized local OT keeps the same convergence logic.

Theorem: topology adds a floor

Over arbitrary connected graphs,

$$\min_{r<R} \Delta_\tau^{(r)} \le \mathcal{O}(1/R) + C\sum_{s<R} \delta_s^2/R.$$

Gossip mixing gives $\delta_{r+1} \le \rho^L(L_\tau\delta_r + \zeta_\tau)$.

Finite-step gossip turns exact descent into inexact descent controlled by consensus error.

For more methodology and theoretical details, please refer to the full paper.

香港中文大學
The Chinese University of Hong Kong

Decentralized WB can learn free supports by aligning moving atom trajectories, not fixed grids.

Scan

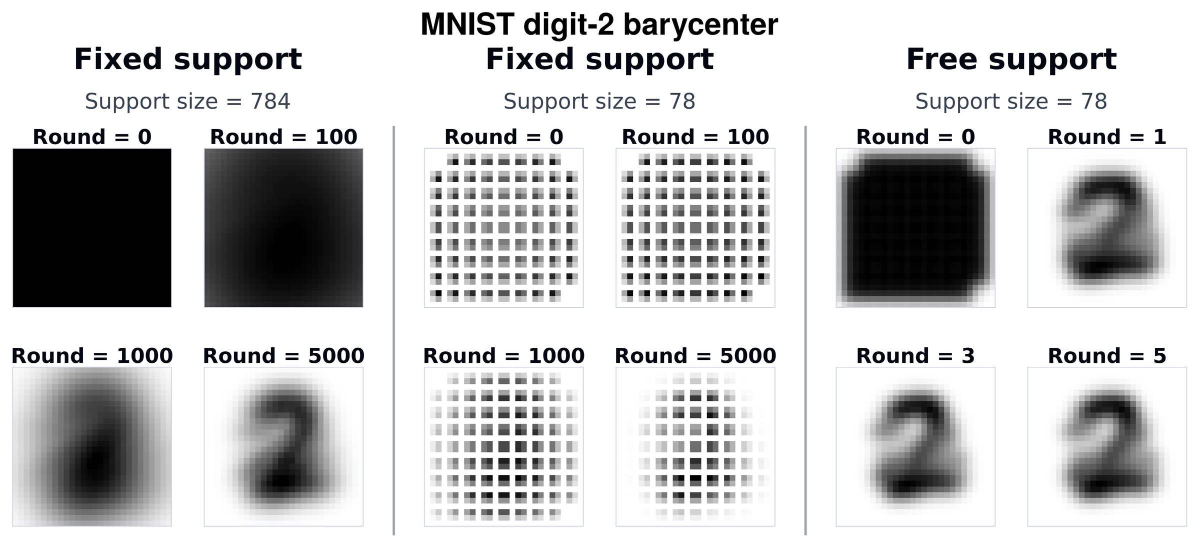
OpenReview profile and contact.

Contact: xinbaoqiao@cuhk.edu.hk

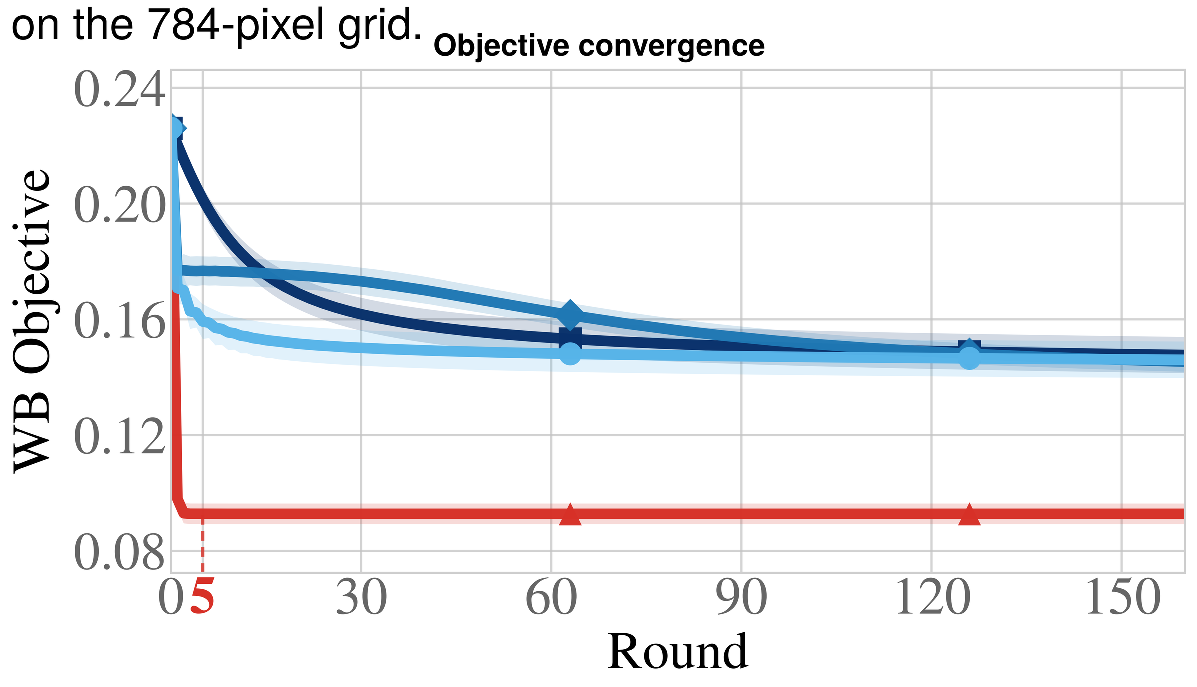
Method

Each agent computes an OT plan from its current barycenter estimate to its local measure, moves atoms along McCann interpolation, and then gossips only the $d \times m$ support matrix. Atom identities are inherited along continuous support trajectories, making averaging of moving supports well-defined.

Primary Evidence



Free support reaches lower evaluated WB objective after 5 rounds while the fixed-support baseline uses 5000 rounds on the 784-pixel grid.



Experimental Takeaway

- MNIST: objective drops from 0.00825 to 0.00682, a 17.34% reduction.
- Resource use falls to 0.00995% compute and 0.0199% communication.
- Truncated Gaussians: objective 0.0886 versus 0.1371–0.1392 for fixed-support decentralized baselines.

References

Agueh and Carlier, SIAM J. Math. Anal. 2011; Cuturi and Doucet, ICML 2014; Dvurechenskii et al., NeurIPS 2018; Kroshnin et al., ICML 2019.

For additional experimental results, please refer to the full paper.

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